I Network and Merchant Incidence

In this appendix we use the original two-sided framework of **?** to show that small reductions in interchange fees have no first-order effects on network profits, even if the pass-through of interchange to rewards is incomplete. One may have an intuition that if networks do not fully pass through interchange fees to rewards, then networks also internalize some profits from reduced interchange. However, the envelope theorem implies that the network's profit change is second-order in the fee change, regardless of how aggressively rewards adjust. This Appendix walks through the details of this argument, and also characterizes the first-order effects on merchants and consumers.

I.A Setup

The monopoly network is a combined entity (network, issuers, acquirers) that sets two per-transaction prices: p_s charged to merchants and p_b charged to cardholders. Rewards correspond to $p_b < 0$.

Payment volumes depend on both consumer and merchant adoption. We model this with a standard two-sided demand structure. A share $D_b(p_b)$ of consumers adopt, and a share $D_s(p_s)$ of merchants accept the card. Card transactions only occur when both consumers want to use the card and merchants accept, and so total payment volume equals $Q = D_b D_s$. Profit is

$$\pi(p_b, p_s) \; = \; (p_b + p_s - c)\, D_b(p_b)\, D_s(p_s). \tag{1}$$

In the unconstrained equilibrium, the network sets optimal prices (p_b^*, p_s^*) to maximize (1). Writing the markup as $\phi \equiv p_b^* + p_s^* - c$, the interior FOCs $\partial\pi/\partial p_b = D_b D_s + \phi\, D_b' D_s = 0$ and $\partial\pi/\partial p_s = D_b D_s + \phi\, D_b D_s' = 0$ yield the slope identities

$$D_b \; = \; -\phi\, D_b', \qquad D_s \; = \; -\phi\, D_s' \qquad \text{at } (p_b^*, p_s^*). \tag{2}$$

I.B The Effects of Interchange Fee Caps on Prices and Profits

When a regulator caps the merchant fee, this lowers consumer rewards but has no first-order effect on network profits.

We first derive the price effect. Suppose a regulator caps $p_s \leq p_s^* - \varepsilon$ for $\varepsilon > 0$ small. The network re-optimizes p_b subject to the constrained merchant price, yielding $p_b^*(\varepsilon)$ defined by the residual buyer-side FOC

$$D_b(p_b^*) \; + \; (p_b^* + p_s^* - \varepsilon - c)\, D_b'(p_b^*) \; = \; 0. \tag{3}$$

Implicitly differentiating (3) at $\varepsilon = 0$ pins down the cardholder-side response,

$$\rho^R \;\equiv\; \frac{dp_b^*}{d\varepsilon}\bigg|_{\varepsilon=0} \;=\; \frac{D_b'(p_b^*)}{2D_b'(p_b^*) + \phi\, D_b''(p_b^*)}. \tag{4}$$

ρ^R measures how much of a merchant-fee cut the network claws back by raising the cardholder price (i.e., cutting rewards). We use the same symbol ρ^R as in Appendix **??**: both measure the fraction of a merchant-fee change that reaches cardholders. There the parameter enters a reduced-form rebate rule; here it is microfounded as the network's profit-maximizing response.

Under log-concave demand, ρ^R is between zero and one. Log-concave demand captures most common demand curves, including linear and isoelastic, and is a standard assumption in the literature on imperfect pass-through (**?**). Thus networks generically only partially offset merchant-fee cuts with lower rewards.

Lemma 1 (Range of ρ^R)**.** *Under log-concavity of cardholder demand at the monopoly point, $\rho^R \in (0,1)$.*

Proof. The denominator in (4) equals $\partial^2\pi/\partial p_b^2 \,/\, D_s$, which is negative by the unconstrained second-order condition. With $D_b' < 0$, this gives $\rho^R > 0$. The upper bound $\rho^R < 1$ requires $D_b' + \phi\, D_b'' < 0$; substituting the FOC identity $D_b = -\phi\, D_b'$ rewrites this as $D_b\, D_b'' \le (D_b')^2$, equivalently $(\log D_b)'' \le 0$. □

Although the cap has first-order effects on consumer prices, the network bears no incidence to first order. The network was already balancing merchant acceptance against cardholder rewards optimally, so a small forced perturbation has no first-order profit effect by the envelope theorem.

Theorem 1 (Zero Network Incidence)**.** *Suppose the network's pre-cap prices satisfy the unconstrained FOCs with negative-definite Hessian and D_b is log-concave at p_b^*. Then $d\pi^*/d\varepsilon\big|_{\varepsilon=0} = 0$.*

Proof. Let $\pi^*(\varepsilon) \equiv \pi(p_b^*(\varepsilon), p_s^* - \varepsilon)$. Differentiating,

$$\frac{d\pi^*}{d\varepsilon} \;=\; \frac{\partial\pi}{\partial p_b}\bigg|_* \cdot \frac{dp_b^*}{d\varepsilon} \;+\; \frac{\partial\pi}{\partial p_s}\bigg|_* \cdot (-1).$$

At $\varepsilon = 0$ both partials vanish by the unconstrained FOCs, so $d\pi^*/d\varepsilon|_{\varepsilon=0} = 0$. □ □

The value of ρ^R—i.e., how aggressively the network re-optimizes rewards in response to the regulation—does not affect first-order network incidence. Although the proof here only applies for small caps, the result holds in estimated structural models of the payment industry. For example **?** finds that even large caps deliver only modest changes in network profits—consistent with the local result here.

I.C First-Order Effects on Each Party

Cardholder and merchant surplus each have two components: an own-side rectangle from the price they face and a cross-side participation externality from changes in the counterparty base. A cardholder with per-transaction valuation b_b who adopts the card earns $b_b - p_b$ on each of D_s accepting merchants, so aggregate cardholder surplus is $CS_b = \bar{S}_b(p_b)\, D_s(p_s)$ where $\bar{S}_b(p_b) \equiv \int_{p_b}^{\infty}(b_b - p_b)\, dF(b_b)$ is the per-merchant inframarginal cardholder surplus. Aggregate merchant surplus is symmetric: $CS_s = \bar{S}_s(p_s)\, D_b(p_b)$ with $\bar{S}_s$ defined analogously.

Differentiating at $\varepsilon = 0$:

$$\Delta_{\text{cardholder}} = \underbrace{-\rho^R\, \varepsilon\, Q^*}_{\text{own-side rectangle}} \quad \underbrace{-\, \varepsilon\, \bar{S}_b\, D_s'}_{\text{new merchants accept}} \quad , \tag{5}$$

$$\Delta_{\text{merchant}} = \underbrace{+\varepsilon\, Q^*}_{\text{own-side rectangle}} \quad \underbrace{+\, \rho^R\, \varepsilon\, \bar{S}_s\, D_b',}_{\text{cardholders exit}} \tag{6}$$

$$\Delta_{\text{network}} = 0, \tag{7}$$

each evaluated at the unconstrained monopoly point with $Q^* = D_b D_s$. Cardholders lose $\rho^R\varepsilon$ per transaction on the inframarginal base but gain access to additional accepting merchants ($D_s' < 0$, so $-\bar{S}_b D_s' > 0$). Merchants gain ε per transaction on the inframarginal base but lose some cardholders to the tighter rewards ($D_b' < 0$, so $\rho^R \bar{S}_s D_b' < 0$). The network change is zero by Theorem 1.

I.D Relating the Rochet-Tirole Calculations to the Main Text

Although the Rochet-Tirole framework provides for many aggregate welfare channels, many of these channels drop out in realistic environments. First, in the U.S. payment market merchant acceptance is largely saturated. Even though fees are high, around 95% of merchants accept cards. Hence aggregate effects from changes in merchant acceptance are likely to be small – i.e. the $D_s^{'}$ terms can be ignored. The main text already imposes this by assuming that all merchants accept all payment methods.

Second, if the main benefit that merchants realize from card acceptance is higher sales, but consumers face a budget constraint, then the externality from higher consumer adoption to greater merchant benefits washes out in aggregate. Hence the $D_b^{'}$ terms for the merchant can be ignored.

Third, if merchants pass through a fraction ρ^P of the fee savings into lower retail prices—the same merchant pass-through parameter as in equation (**??**) of Appendix **??**—then part of the merchant gains are transferred to consumers. Because Rochet–Tirole takes merchant pricing as exogenous, ρ^P is a primitive in this appendix rather than a derived object. Thus the consumer benefit should add a $\rho^P \varepsilon Q^*$ term from the merchant side to the consumer side. The net result is the following aggregate welfare effects:

$$
\begin{aligned}
\Delta_{\text{cardholder}} &= (\rho^P - \rho^R)\, \varepsilon\, Q^* \\
\Delta_{\text{merchant}} &= (1 - \rho^P)\, \varepsilon\, Q^* \\
\Delta_{\text{network}} &= 0.
\end{aligned}
$$

Summing the three party effects, the cap raises total surplus by $(1 - \rho^R)\, \varepsilon\, Q^*$. This residual is the deadweight loss the cap recovers: ρ^P only redistributes the merchant gains to cardholders, while ρ^R governs how much of the cap genuinely lowers the consumer price of card use rather than being clawed back through lower rewards. Thus, although networks bear no first-order incidence from capping interchange, the rewards pass-through ρ^R remains first-order for both consumer welfare and aggregate efficiency.